

Babalola John Abidemi

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EDUCATION

Ladoke Akintola University of Technology (LAUTECH), Ogbomoso, Oyo State
B.Tech in Electronic and Electrical Engineering CGPA: **4.69/5.0 (3.8/4.0) WES**

Dec. 2021 - Aug 2026

Software Skills: Python, C++, Robot Operating System (ROS2- SLAM, NAV2, Path Planning, Teleoperation, ROS - AI Integration), Agentic AI, MATLAB/Simulink, Pytorch.

Hardware Skills: Embedded System, Arduino Programming, CAD/CAM Design (Fusion360), Addictive and Subtractive Manufacturing, hardware programming (Raspberry pi, Arduino, ESP32), Laser Cutting/Engraving.

Research Interest: Robot Perception and Autonomous System, Human Robotic Interaction (HRI), Space and Field Robotics, Embodied AI and Agent-based Robot Architecture, Sensor Fusion and Edge AI for Autonomous Systems, Agricultural Robotics and Resourced Constrained Robot.

RESEARCH AND WORK EXPERIENCE

Undergraduate Researcher

Unilag Design Studio University of Lagos, Lagos

May. 2025 - Oct. 2025

- Designed an AI-powered pick-and-place robot that translates natural language commands into autonomous manipulation in confined spaces using ROS 2 (MoveIt 2), computer vision and agentic AI.
- Developed **VisionLock**, a high-performance embedded AI security system leveraging computer vision-based identity recognition for autonomous access control.
- Designed and implemented **CourierX**, a digital-twin enabled, AI-powered autonomous delivery robot integrating ROS, LiDAR-based perception, SLAM and autonomous navigation for safe and reliable last-mile delivery.
- Developed a comprehensive, beginner-friendly tutorial covering Convolutional Neural Networks (CNNs), PyTorch and computer vision fundamentals.
- Architected an end-to-end AI system integrating research automation, web scraping pipelines, intelligent task orchestration and AI-powered home automation services within a unified framework

Student Industrial Work Experience Scheme

Unilag Design Studio University of Lagos, Lagos

Mar. 2025 - May. 2025

- Received hands-on training in embedded systems, Arduino programming and IoT architectures applied to robotics and intelligent systems.
- Trained in project management methodologies and the art of producing clear, structured technical documentation.
- Led a team of five to design and develop a ROS-based fully autonomous mobile robot (AMR) for intelligent navigation and learning applications.

Embedded Systems Intern

Elord Technology, Ogbomosho, Lagos

May. 2024 - May. 2025

- Implemented infrared sensor configurations for closed-loop solar tracking systems, enabling automated panel alignment and improved energy capture.
- Developed an automated fire detection and alert system for industrial environments, integrating real-time sensing with remote emergency notification.
- Developed **Attendify**, an RFID-enabled smart attendance system integrating embedded hardware with cloud-based data logging and automation via Google Sheets.

Student Work Experience Program

Ladoke Akintola University of Technology, Electronic/Electrical Department,

Feb. 2024 - May. 2024

- Received foundational training in basic electronics and electrical circuitry.
- Trained in writing comprehensive and well-structured technical project reports.

EXTRA-CURRICULAR/LEADERSHIP EXPERIENCE

Institute of Electrical and Electronic Engineering LAUTECH Student Branch | Ogbomosho.

Project Coordinator |

Mar. 2025 - Present

- Led and facilitated an impactful 3-day embedded systems bootcamp for undergraduates, introducing Arduino programming and the application of embedded systems in robotics.
- Hosted a technical session on bridging the gap between semiconductor technologies and real-world industrial applications.
- Led and coordinated student projects and participated in writing technical grant applications for robotics and environmental sustainability initiatives.

Low Budget Engineer | Ibadan, Oyo State.

Exhibition and STEM Educator |

July. 2025 - Present

- Coordinated and supported the delivery of STEM education programs and technical exhibitions, promoting hands-on learning and innovation.
- Led and contributed to the development of impactful, solution-oriented engineering projects addressing real-world challenges.

BLHCF LAUTECH | Ogbomosho, Oyo State.

Academic Coordinator |

Feb. 2023 - Present

- Coordinated and managed impactful academic tutorials to enhance student learning and engagement.
- Provided academic guidance and career mentorship to undergraduate students, supporting their development and success.

AWARDS AND RECOGNITIONS

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|--|-----------------------------------|
| ● Dean's List Award - Perfect GPA (5.00) | <i>2023/2024 academic session</i> |
| ● Top 1% - Electronic Engineering | <i>2023/2024 Academic Session</i> |
| ● Top 1% - Control Systems and Engineering | <i>2024/2025 Academic Session</i> |
| ● Best Intern - Embedded Systems and AI | <i>Unilag Design Studio, 2025</i> |
| ● Best Intern - Embedded Systems, Computer Vision and ROS | <i>Elord Technology, 2025</i> |
| ● Oluwasedago Foundation - Scholarship of Excellence | <i>2025</i> |
| ● Co-founder - IEEE LAUTECH Student Branch | <i>2025</i> |

ACADEMIC CONFERENCES/SEMINAR

- Attended African Engineering and Technology Network, Conference titled “**Energy Sustainability Panacea For Africa Energy crisis**” at University of Lagos, Nigeria, 30th April, 2025
- Attended African Engineering and Technology Network, Conference titled “**Empowering the Next Generation of African Research Leaders in Digital World**” at University of Lagos, Nigeria, 28th May, 2025

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Advanced Computer Vision and DeepLearning, Udemy *2025*

- Advanced image processing and deep learning model training using PyTorch and TensorFlow, including CNN theory and applications.
- Practical experience with transfer learning, fine-tuning pretrained models, object detection frameworks (YOLOv8, Faster R-CNN) and data preprocessing pipelines.

Mastering Pytorch: 100 project Bootcamp Training, Udemy *2025*

- Completed 100 projects covering CNN model training, transfer learning and LLM applications using PyTorch.

Edge Avoiding Robot Using ROS, Mechamind *2025*

- Learned ROS fundamentals, including writing nodes, publishing/subscribing, URDF creation, robot simulation in Gazebo and sensor-based edge avoidance implementation.

AI Agents Using Langchain and Langraph, Huggingface *2025*

- Learned to integrate AI agents into embedded systems, computer vision pipelines and ROS for intelligent autonomous system development.

VOLUNTEERING AND PROFESSIONAL MEMBERSHIP

Volunteering & Coordination Roles

- **Project Coordinator** - IEEE LAUTECH Student Branch *2025 - Present*
- **Executive** - Robotics and Artificial Intelligence Nigeria (RAIN) *2025 - Present*
- **Academic Coordinator** - BLHCF LAUTECH *2025 - Present*
- **Volunteer** - Heal Me Foundation *2024 - Present*
- Low Budget Engineer *2025 - Present*

Professional Memberships

- Member - Society of Petroleum Engineers (SPE), | ID: 5743743 *2024 - Present*
- Member - International Association of Engineers (IAENG) *2024 - Present*
- Member - Institute of Electrical and Electronic Engineers | ID: 100929513 *2025 - Present*
- Member - Robotics and Artificial Intelligent Nigeria(RAIN) *2025 - Present*